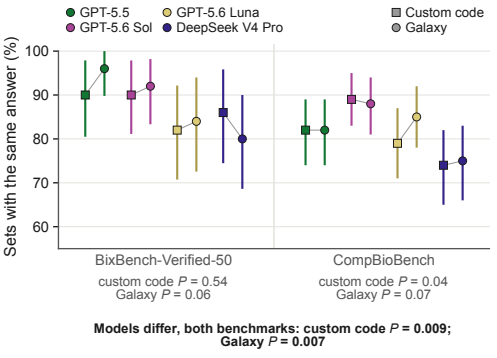
a What each model ran in Galaxy

Galaxy runs with at least one completed job in each family (%); UDTs were not offered on IWC

		BixBench				CompBio				IWC			
Data handling	Tables and text	10	38	51	33	17	28	43	33	0	33	33	37
	Format conversion and upload	19	42	57	23	10	16	31	22	7	26	22	7
	Inspection and quality control	2	2	2	0	10	22	26	21	0	15	15	11
	Data retrieval	0	0	0	0	0	2	6	4	11	15	11	15
Scientific methods	Read processing and alignment	2	2	2	2	8	16	18	15	30	41	33	30
	Variant calling and annotation	0	0	0	0	3	6	10	7	0	0	0	0
	Expression and differential testing	3	6	9	5	2	3	4	5	22	19	22	11
	Single-cell and spatial analysis	0	0	0	0	1	4	5	7	4	4	0	11
	Genomic intervals and sequence features	2	2	2	2	8	15	26	23	4	7	7	15
	Phylogenetics	12	16	21	17	0	0	0	0	0	0	0	0
	Sequence search, taxonomy and assembly	0	1	1	1	4	5	8	7	11	22	19	7
	Peak calling and epigenomics	0	0	0	0	1	1	2	2	4	11	11	4
	Statistics, machine learning and enrichment	22	35	34	19	2	4	9	9	0	0	0	0
	Other methods	0	0	0	0	0	0	0	0	4	11	7	11
	UDT (agent-written code)	66	35	39	9	56	75	41	35	0	0	0	0
		GPT-5.5	Sol	Luna	DeepSeek	GPT-5.5	Sol	Luna	DeepSeek	GPT-5.5	Sol	Luna	DeepSeek

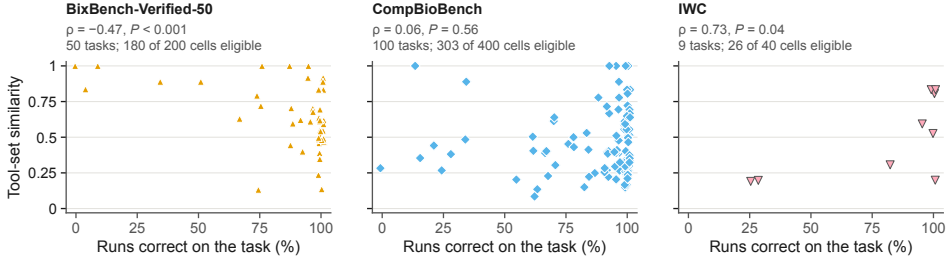
b Same answer in all three runs

Replicate sets (one task, model and condition)



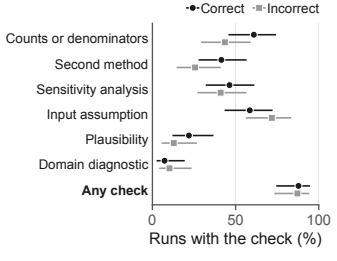
c Tool-set similarity and task accuracy

Galaxy; 1 = the same set of tools in all three runs (any UDT counted as one item; order, versions and parameters ignored). Models compared on the same task: Spearman ρ = 0.03, *P* = 0.57



d Verification checks

80 coded runs (coder blind to the grade); 95% intervals



e Replicate outcomes by held-out task difficulty

BixBench-Verified-50 and CompBioBench, both conditions; difficulty from the task's other 21 runs, so a set's own runs never define it

